

**SAFETY DATA SHEET**

Creation Date 21-Jul-2014

Revision Date 24-Dec-2021

Revision Number 6

**1. Identification**

**Product Name** Diethylaminosulfur trifluoride

**Cat No. :** AC216110000; AC216110050; AC216110250; AC216111000

**CAS No** 38078-09-0  
**Synonyms** DAST

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |              |
|--------------------------------------------------|--------------|
| Flammable liquids                                | Category 3   |
| Acute oral toxicity                              | Category 4   |
| Acute dermal toxicity                            | Category 4   |
| Acute Inhalation Toxicity - Vapors               | Category 4   |
| Skin Corrosion/Irritation                        | Category 1 A |
| Serious Eye Damage/Eye Irritation                | Category 1   |
| Specific target organ toxicity (single exposure) | Category 3   |
| Target Organs - Respiratory system.              |              |

**Label Elements**

**Signal Word**  
Danger

**Hazard Statements**

Flammable liquid and vapor  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep cool

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Risk of explosion if heated under confinement  
Reacts violently with water

### 3. Composition/Information on Ingredients

| Component                                       | CAS No     | Weight % |
|-------------------------------------------------|------------|----------|
| Sulfur, (N-ethylethanaminato)trifluoro-, (T-4)- | 38078-09-0 | 95       |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                    |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                    |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                              |
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.                                                       |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                |

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER                                                                                                    |
| <b>Flash Point</b>                      | 23 °C / 73.4 °F                                                                                                     |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Hydrogen cyanide (hydrocyanic acid). Gaseous hydrogen fluoride (HF). Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
2

**Instability**  
3

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                                                                                         |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.                                          |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Corrosives area. Keep away from water or moist air. Keep away from heat, sparks and flame. Store in freezer. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Acids. Water. Strong oxidizing agents. Strong bases. Alcohols.                                                                                                                                                              |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                                              | ACGIH TLV                  | OSHA PEL                             | NIOSH IDLH                  | Mexico OEL (TWA)           |
|--------------------------------------------------------|----------------------------|--------------------------------------|-----------------------------|----------------------------|
| Sulfur,<br>(N-ethylethanaminato)trifluor<br>o-, (T-4)- | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
OSHA - Occupational Safety and Health Administration

|                             |                                                                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b> | Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Personal Protective Equipment

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Tight sealing safety goggles. Face protection shield.                                                                                                                                                                                             |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                            |                                    |
|----------------------------|------------------------------------|
| <b>Physical State</b>      | Liquid                             |
| <b>Appearance</b>          | Dark yellow                        |
| <b>Odor</b>                | No information available           |
| <b>Odor Threshold</b>      | No information available           |
| <b>pH</b>                  | No information available           |
| <b>Melting Point/Range</b> | No data available                  |
| <b>Boiling Point/Range</b> | 30 - 32 °C / 86 - 89.6 °F @ 3 mmHg |

|                                        |                                  |
|----------------------------------------|----------------------------------|
| Flash Point                            | 23 °C / 73.4 °F                  |
| Evaporation Rate                       | No information available         |
| Flammability (solid,gas)               | Not applicable                   |
| Flammability or explosive limits       |                                  |
| Upper                                  | No data available                |
| Lower                                  | No data available                |
| Vapor Pressure                         | No information available         |
| Vapor Density                          | No information available         |
| Specific Gravity                       | 1.316                            |
| Solubility                             | Decomposes in contact with water |
| Partition coefficient; n-octanol/water | No data available                |
| Autoignition Temperature               | No information available         |
| Decomposition Temperature              | No information available         |
| Viscosity                              | No information available         |
| Molecular Formula                      | C4 H10 F3 N S                    |
| Molecular Weight                       | 161.18                           |

## 10. Stability and reactivity

|                                  |                                                                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                                                                                                          |
| Stability                        | Moisture sensitive.                                                                                                                                                                                                                          |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition. Temperatures above 60°C.                                                         |
| Incompatible Materials           | Acids, Water, Strong oxidizing agents, Strong bases, Alcohols                                                                                                                                                                                |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Hydrogen cyanide (hydrocyanic acid), Gaseous hydrogen fluoride (HF), Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                                                                                                     |
| Hazardous Reactions              | Reacts violently with water.                                                                                                                                                                                                                 |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                              | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Sulfur,<br>(N-ethylethanaminato)t<br>rifluoro-, (T-4)- | 38078-09-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                      |                           |
|----------------------|---------------------------|
| Reproductive Effects | No information available. |
|----------------------|---------------------------|

|                                                                  |                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                                     | No information available.                                                                                                                                                                                                                                                                                                                       |
| <b>Teratogenicity</b>                                            | No information available.                                                                                                                                                                                                                                                                                                                       |
| <b>STOT - single exposure</b><br><b>STOT - repeated exposure</b> | Respiratory system<br>None known                                                                                                                                                                                                                                                                                                                |
| <b>Aspiration hazard</b>                                         | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Symptoms / effects, both acute and delayed</b>                | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>                           | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Other Adverse Effects</b>                                     | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                  |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                                 |
|--------------------------------------|-----------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available.         |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                       |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its volatility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Technical Name</b>          | (DIETHYLAMINOSULFUR TRIFLUORIDE)    |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | I                                   |

### TDG

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | I                                   |

### IATA

|                                |                                      |
|--------------------------------|--------------------------------------|
| <b>UN-No</b>                   | UN2920                               |
| <b>Proper Shipping Name</b>    | CORROSIVE LIQUID, FLAMMABLE, N.O.S.* |
| <b>Hazard Class</b>            | 8                                    |
| <b>Subsidiary Hazard Class</b> | 3                                    |
| <b>Packing Group</b>           | I                                    |

### IMDG/IMO

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | I                                   |

## 15. Regulatory information

### United States of America Inventory

| Component                                             | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-------------------------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Sulfur,<br>(N-ethylethanaminato)trifluoro-,<br>(T-4)- | 38078-09-0 | -    | -                                             | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                                             | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-------------------------------------------------------|------------|-----|------|-----------|-------|------|------|------|-------|------|
| Sulfur,<br>(N-ethylethanaminato)trifluoro-,<br>(T-4)- | 38078-09-0 | X   | -    | 253-771-2 | X     | -    |      | -    | -     | -    |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Health Administration Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Component                                          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------------------------------|---------------|------------|--------------|----------|--------------|
| Sulfur,<br>(N-ethylethanaminato)trifluoro-, (T-4)- | -             | X          | -            | -        | X            |

### U.S. Department of Transportation

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland** This product does not contain any DHS chemicals.

## Security

Other International Regulations

Mexico - Grade

Serious risk, Grade 3

Authorisation/Restrictions according to EU REACH

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                       | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Sulfur, (N-ethylethanaminato)trifluoro-, (T-4)- | 38078-09-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                       | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Sulfur, (N-ethylethanaminato)trifluoro-, (T-4)- | 38078-09-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

## Prepared By

Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

## Creation Date

21-Jul-2014

## Revision Date

24-Dec-2021

## Print Date

24-Dec-2021

## Revision Summary

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**